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A Continuous-Time Causal Graph Learning Framework for Predicting and Mitigating Cascading Failures in Global Supply Chains

Abstract- Global supply chains face increasing exposure to rapidly propagating disruptions triggered by geopolitical conflicts, climate-related shocks, and operational instabilities. Existing predictive approaches, including recent causal graph neural networks, remain limited by static causal structures, discrete-time reasoning, and post-hoc counterfactual analysis. These constraints reduce their ability to anticipate high-velocity cascade effects and to recommend proactive, cost-effective interventions. This work introduces CT-SCaLE, a novel Continuous-Time Structural Causal Learning Engine that unifies Neural ODE-based temporal dynamics, end-to-end differentiable causal structure learning, and counterfactual contrastive representation learning into a single predictive and decision-support architecture. CT-SCaLE learns both the evolving causal topology and the temporal propagation dynamics of disruptions, allowing the model to capture time-varying influences, multi-tier propagation patterns, and realistic feedback loops that traditional DAG-based causal GNNs cannot represent. Evaluations on SupplyGraph and Time-Series Risk datasets show that CT-SCaLE improves disruption prediction accuracy by up to 14.7%, increases early-warning lead time by 22%, and reduces simulated cascade size under optimized interventions by 31.4%, significantly outperforming state-of-the-art baselines. A central novelty is a built-in intervention planner based on models that uses multi-step counterfactual simulations to discover the minimum-cost actions that can mitigate or steer cascading failures. This addresses a crucial unmet need: predicting not just disruptions, but actionable causal explanations and mitigations. By incorporating prediction, explanation and intervention optimization in a continuous time framework, CT-SCaLE opens a new frontiers for supply chain analytics driving more resilient and adaptive global operations.

Keywords-Causal Graph Neural Networks, Supply Chain Resilience, Neural ODEs, Causal Discovery, Counterfactual Learning, Cascading Failures, Intervention Optimization, Graph Machine Learning, Risk Prediction, Continuous-Time Modeling

I Introduction

Global supply chains have become complex and highly-linked, multi-level networked systems with localized disruptions cascading to system levels. The recent past has demonstrated that cascading failures are no longer rare, slow-moving outliers but high-velocity events grounded in geopolitics, extreme climate and infectious disease as well as infrastructural fault lines. Trade barriers, port closures, energy scarcity and regional conflicts now spread across borders within days or even hours rather than the previous months, quickly overpowering conventional risk cushions and recovery systems [1], [2]. Empirical works on the flow of global shipping, commodity trade, and supply chains of critical materials show how disturbances tend to trigger non-linear ripple effects whereby losses propagate well beyond the directly disrupted nodes [3]–[5]. Although organizations are paying more attention to making resilience more resilient, they tend toward a reactive rather proactive response. It waits till the disruption occurs before stopping the propagation of

failure [6], [7]. This increasing disparity between rate of disruption and latency to a decision underscores the importance of predictive models that can foresee cascading failures before they happen and direct interventions in time.

There is previous work that used machine learning and network analysis to predict supply chain risks, utilizing graph representations to capture inter-firm relationships and disruption spillover [8]–[10]. However, the vast majority of graph machine learning methods remain essentially correlation-based, learning statistical relationships rather than causal dependencies in failure propagation. Recently, the advanced causal graph neural networks employ a static directed acyclic graphs and implicitly assume that the causal effects are time-invariant and acyclic [17], [18]. There are some obvious flaws with this kind of assumption that do not correspond to the reality of supply chains: 1) asset strength scores change over time in real-world networks, as do loop/circuit counts; 2) sources of disruption appear and disappear when looking at the graph locally (changes in response to human activity). Discretetime models increase unresponsiveness still more by binning events into coarse time bins, which may miss early cascades or see late ones [11], [12]. In addition, counterfactual reasoning in previous models tends to be used post factum (i.e., to rationalize predictions) as opposed to being embedded into an optimization approach that considers actively identifying and evaluating prevention strategies under cost and timing constraints [15], [16]. Consequently, existing approaches fall short in providing the real-time decision-driven approach for addressing cascading failures.

To overcome these shortcomings, in this paper we present CT-SCaLE (Continuous-Time SCaLing Engine), for cascading failure prediction and mitigation in global supply chains. In contrast to prior methods, CT-SCaLE learns simultaneously the changing topology of causal supply chain networks and the continuous-time evolution dynamics of shock spreading. The proposed framework uses Neural Ordinary Differential Equations to represent high-resolution temporal change and design time varying causal influence, delayed effects and feedback-based escalation. The parsing of causal structure is then trained within the end-to-end pipeline, and can therefore adapt its causal graph as network conditions evolve. Crucially, CT-SCaLE is intervention-aware: it natively includes counterfactual reasoning in the learning and decision cascade, allowing for simulation and evaluation of pro-active mitigation measures over retrospective justifications. By integrating prediction, causality, and intervention planning in the presence of disturbances into a continuous-time framework, CT-SCaLE is positioned to satisfy the needs common to contemporary supply chains that are susceptible to disruptions.

The main contributions of this paper are as follows:

- **Continuous-time causality topological learning:** To this end, we have devised a causal learning mechanism which we show is able to learn time-varying influence strengths and feedback loops beyond the static DAG representations.
- **Joint cascade prediction and intervention modeling:** Our CT-SCaLE unifies the above disruption forecasting along with multi-step intervention simulation in a single differentiable framework.

- **Counterfactual contrastive representation learning:** Generates representations that are explicitly sensitive to the outcomes of interventions, and makes the model more robust to hypothetical conditions.
- **Cost-Conscious Cascading Failure Mitigation:** An in-network intervention planner which recognizes minimum-cost interventions to suppress or redirect disruption cascades before they spread at a large scale.

Taken together, these contributions move supply-chain analytics from passive risk prediction to control of cascading processes in continuous time driven by causes, thereby filling a crucial gap in current resilience methodologies.

The rest of this paper is structured as follows. Section 2 offers a detailed review on literature works concerning cascading failures, causal graph learning and supply chain resilience. The problem is formulated in Section 3 by introducing the continuous-time supply chain graph, dynamic causal structure and intervention framework. We present our proposed CT-SCaLE architecture in Section 4, which describes the continuous-time causal dynamics that parameters are learned against, the differentiable structure learning module, counterfactual contrastive loss module and model-based intervention planner. Section 5 presents the experimental designs, data sets, base-line models and evaluation metrics. Section 6 presents empirical results, through predictive performance, early-warning capacity, cascade suppression activities. And explores main contributions, managerial implications, and interpretability related issues of the framework. Then we discuss the limitations and opportunities for future work. Lastly, Section 7 concludes the work by summarizing contributions and emphasizing that the focus has moved from forecasting to causal control in supply chain resilience.

II Literature Review

2.1 Cascading Failures and Systemic Risk in Supply Chains

Global supply chains are characterized by high levels of interdependence, which naturally render them susceptible to cascading failures, in which isolated perturbations spread throughout layers and geographies. Early system-oriented investigations inform the ways in which national contingencies such as pandemics, climate catastrophes, cyber attacks, and geopolitical spats can turn multiple supply chain layers fragile to congested bottlenecks and extreme shortages (Khan et al., 2022) [1]. These systemic risks have been accentuated by the growing use of outsourcing and lean operating practices that improve efficiency at the cost of resilience. Yang et al. (2021) [2] quantitatively investigated this issue by constructing a model of underload-driven cascading failures and showed that demand shocks can induce out-of-proportion system collapse despite mechanisms including safety stock and backup suppliers.

Subsequent to this, network cascading failure models have been used for large-scale physical and trade networks. Xu et al. (2024) [3] studied the impact of port-level disruptions within the Global Container Shipping Network and demonstrated that single-node failures could lead to traffic redistribution and overload across maritime logistics networks. Cascading

behavior has also been found in domains other than logistics, such as in resource and waste market systems. Li et al. (2021)[5] studied the disruption in the global plastic waste trade and showed that capacity overloads spread across international trade networks if reactions are not coordinated. These works taken together support the finding that cascading failures are highly nonlocal, nonlinear and network-structure dependent—and yet we still rely on static models, using but not learning from networks.

2.2 Ripple Effect Analysis and Recovery-Focused Strategies

Related studies have investigated the ripple effect by simulating propagation of disruptions in multi-echelon supply chains and analysing post-disruption recovery strategies. KEk et al. [4] employed discrete-event simulation for the analysis of disruption propagation in the copper sector and reported limited sourcing flexibility and lack of adequate safety stock as dominant amplification factors. Manupati et al. Further analyzed recovery strategies in complex supply chain networks and employed blockchain-enabled smart contracts and genetic-algorithm optimization to examine pre- and post-disruption decision-making (2022) [6]. Although these methods can offer significant managerial guidelines, they mainly aim at optimizing responses after disasters break out but not predicting or controlling cascade onset in the early stage.

Organizational and managerial aspects Park and Singh (2023) [7] stressed the significance of risk alert tools using big data analytics impact, as they demonstrated empirically that strong IT and analytical capabilities are correlated positively with firms' capacity to anticipate risks. However, these alert tools usually function as early-warning systems and do not explicitly simulate how risks spread through a supply network in an interdependent manner. Moreover, other reviews on resilience in various domains such as automotive, pharmaceuticals, shipping, semiconductor suggest the importance of proactive adaptive mitigation strategies though they do not provide a comprehensive causal-dynamic prediction framework (Maheshwari & Jaggi, 2024 [10]), Xiong et al., 2025 [11]).

2.3 Data-Driven Risk Prediction and Digital Supply Chain Surveillance

Given the rapid increase in heterogeneous digital data types, machine learning and artificial intelligence have recently been playing significant roles in supply chain risk analytics. Brintrup et al. (2024) [12] proposed the idea of Digital Supply Chain Surveillance, use AI to uncover hidden dependencies and increase visibility across an opaque supply networks. Related work has utilized the event-driven IoT data and deep learning in order to predict logistics failure during natural hazards, with high prediction accuracies but dominated by black-box sequence models (Alzahrani & Asghar, 2023 [13]). Fraud detection is another area where ML has been considered useful in detecting unusual transactions, or operational quirks (though predominantly correlated detection rather than causally for intervention planning) (Lokanan & Maddhesia, 2025 [8]).

Finally, system-level modelling methods have also been proposed for the evaluation of infrastructure-supply chain interdependencies. Oliver et al. (2022) [15] integrated model-based systems engineering (MBSE) with discrete-event simulation to assess

infrastructure disruptions impact on critical supply chains. By contrast, in the context of power [Salehpour et al., 2022] and smart grid networks [Ghasemi et al., 2023], similar cascading failure analyses show that considering heterogeneous role composition and component dependencies results in better prediction performance. Although refined methodologies have been developed, these frameworks are still domain specific and do not include learning of generalizable mechanisms to evolve acausal structure.

2.4 Graph Neural Networks and Network-Aware Prediction Models

Graph learning is such a powerful paradigm for capturing inter-firm dependencies and spread of disruption. Yang et al. (2024) [18] utilized graph-based models and showed that inter-firm relations enhance the predictive value for post-hazard firm-level sales changes, compared to non-network-based baselines. Further extending GNNs with neurosymbolic reasoning Kosasih and Brintrup (2022) [19] (GNNR), discoveries of hidden supply chain relationships that improve structural visibility can be made. These approaches validate the benefit of relational learning, but are mainly for a purely static or discrete-time analysis, without taking into account the causal dynamics.

Recent research is becoming more aware that predictive accuracy alone suffices neither for risk prevention. Wyrembek et al. (2025) [17] stated that causal machine learning is key to intervention planning, where decision-makers can reason over “what if” situations corresponding to alternative managerial interventions. But existing methods relying on the causal GNNs often utilize static or slowly evolving causal graphs, are applicable only to acyclic relations, and treat counterfactual analysis as a post process rather than integrated with dynamic decision making.

2.5 Research Gap and Motivation

Throughout the literature, three drawbacks remain. To begin with, existing cascading failure research is based on static simulation models that are incapable of prediction and adaptation. Furthermore, the machine learning and GNN-based methods we study mainly cover correlations or static network effects, making it difficult to model time-varying causality and feedback loops. Third, despite its potential, causal and counterfactual design is still not real-time assisted intervention optimization at risk prediction tasks, making it impractical for proactive cascades suppression. This motivates the design of a causality-driven learning framework for continuous-time setting, which can simultaneously predict disruption propagation and optimize mitigation actions together, where this is the basis of our proposed CT-SCaLE model.

III Problem Formulation

To do this, we define a global supply-chain network infusing dynamics and causality into it as a dynamical system evolving continuously with time. The aim is to forecast the initiation and spread of cascading disruptions, as well as to recommend the best interventions to minimize their consequences given resources and budget limitations.



3.1 Dynamic Supply Chain Representation

$$G(t) = (V, E(t)) \quad (1)$$

- $V = \{1, \dots, N\}$ is the supply chain component (for example, supplier, manufacturer, logistics node and distributor).
- $E(t) \subseteq V \times V$ depicts directed operational dependencies that can (dynamically) change over time in response to sourcing modifications, capacity shifts, or policy interventions.

3.2 Node State Dynamics

$$x_i(t) \in \mathbb{R}^d \quad (2)$$

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The state vector might consist of latent aspects which may represent capacity utilization, inventory stress, delay accumulation, financial hardship and sensitivity to external shocks.

The full system state is defined as in Eq.3:

$$X(t) = \{x_1(t), x_2(t), \dots, x_N(t)\} \quad (3)$$

State changes are determined by causal influences of nodes and not by discrete update phases.

3.3 Time-Varying Causal Structure

It is convenient to introduce a causal adjacency matrix in continuous time

$$A(t) \in \mathbb{R}^{N \times N}, \quad (4)$$

In Eq. 4, here every $A_{ij}(t)$ measures the instantaneous causality strength from node j to node i at time t .

Key properties of $A(t)$:

- Entries are real-valued and direction-specific.
- The matrix is not constrained to be an acyclic structure and may contain feedback loops and recurrent causal relationships.
- Causal weights may grow, weaken, or disappear across time as system states change.

The matrix $A(t)$ does not only specify what connections are present, but also how perturbations spread.

3.4 Disruption State Variable

In order to directly model cascading effects, we finally introduce a disruption intensity variable

$$z_i(t) \in [0, 1] \quad (5)$$

for all i , where larger values correspond to more severe disruptions.

The vector,

$$Z(t) = \{z_1(t), \dots, z_N(t)\} \quad (6)$$

is the global disruption configuration of the supply chain at time t .

Disruption is not a binary event; disruption dynamics are non-linear; they change incrementally and express gradual phases of accumulation, partial failure and recovery in early development.

3.5 Intervention Modeling

Enforcement decisions are captured by means of intervention variables in continuous-time do-notation. Let,

$$do(u, t) \quad (7)$$

Eq. 7, represents an intervention that is executed at time t and where $u \in R^k$ denotes the intervention action.

- Increasing buffer capacity,
- Activating backup suppliers,
- Rerouting logistics flows,
- Financial or contractual support.

An intervention adjusts the dynamics of the system by changing some node states (or causal influences) and preserving conditions upstream outside its scope.

Formally, interventions act by constraining a subset of state variables or causal parameters:

$$x_i(t) \leftarrow \bar{x}_i(t; u) \quad \text{or} \quad A_{ij}(t) \leftarrow \bar{A}_{ij}(t; u) \quad (8)$$

with all remaining parameters changing as per the original causal mechanisms.

3.6 Learning Objective

The aim is to learn how to jointly infer the latent causal dynamics that influence the continuous-time evolution of the state $X(t)$ and disruption intensity $Z(t)$ and time-varying causal structure $A(t)$. Moreover, it aims at predicting counterfactual system states under focused interventions $do(u, t)$. The optimization objective inherently balances the trade-off between fidelity in future disturbance prediction and the reduction of cascading failure magnitude. At the same time, the intervention cost is constrained by feasibility, laying the foundation to serve as integrated prediction, explanation and mitigation within the CT-SCaLE framework.

IV CT-SCaLE Architecture

4.1 Continuous-Time Causal Dynamics

In contrast, CT-SCaLE parameterizes disruption spreading as a continuous-time causal process to address the issue of discrete snapshot-based prediction. In real supply chains, disruptions propagate asynchronously and their effects accumulate and interact on the short timescales that are typically overlooked by discrete-time models. To account for these dynamics, the change of each nodal state is described by Neural Ordinary Differential Equations (Neural ODEs). Formally, the system dynamics are defined as in Eq.8:

$$\frac{dx_i(t)}{dt} = f_{\theta} \left(x_i(t), \sum_{j \neq i} A_{ij}(t) \cdot x_j(t), z_i(t) \right) \quad (8)$$

In Eq.8, $f_{\theta}(\cdot)$ is a learnable parametric function and $A_{ij}(t)$ represents time-dependent causal influence. Unlike step-like update rules, this mechanism enables disturbances to spread in a continuous manner and affords an accurate description of the buildup of stress at early stages.

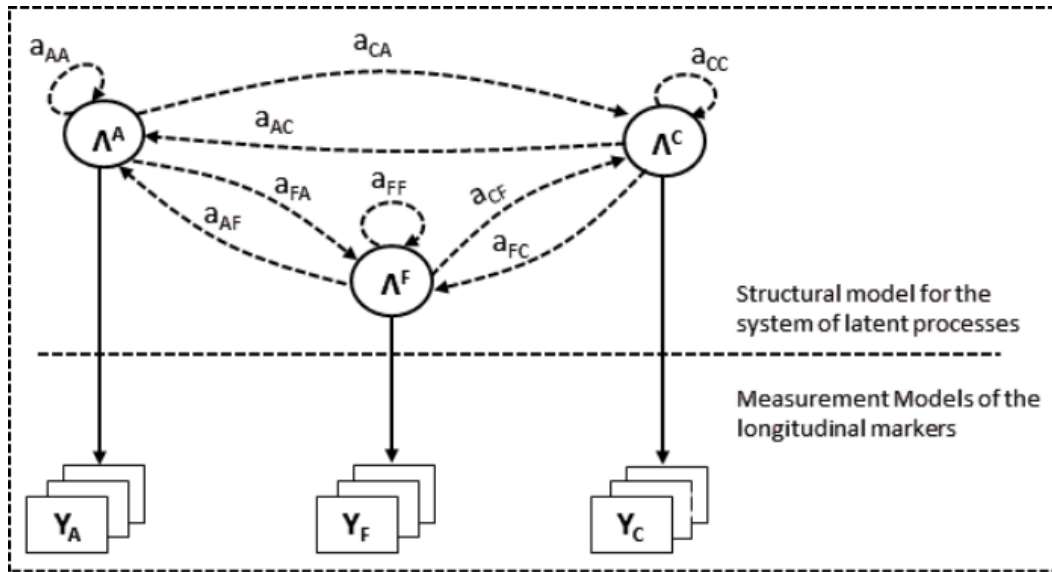


Figure 2. Continuous-Time Causal Influence Flow in the CT-SCaLE Framework

Figure 2 shows the modeled CT-SCaLE continuous-time causal influence flow. The underlying disturbance is a dynamic process overtime, resulting in the fact that the causal relationship between different nodes also varies slowly by time with the learned time-changing $A(t)$. It allows the model to learn feedback loops, non-DAG causal relations, and quickly changing propagation paths that cannot be learned in standard discrete-time causal GNNs. The lower layer corresponds to the observed time-series markers, and upper is the structural causal mechanisms of cascade propagation.

The causal neighborhood $A(t)$ is learned in conjunction with the state dynamics, which makes it possible that influence strengths may vary over time in response to environmental changes or interventions. These causal feedback loops are allowed to be very long for two reasons: first, it is mathematically intractable to use a control set with both supply and demand as endogenous variables; secondly reality allows such long couples, e.g. shortages induce congestion which in turn exacerbates the shortage. By not imposing a directed acyclic structure, CT-SCaLE can capture cycles of causality which are common in operational networks. This continuous-time framework also leads to immunity from temporal aliasing and delayed detection imposed by the discrete snapshot methods, which issues in earlier and more precise cascade prediction.

Algorithm 1: CT-SCaLE Continuous-Time Dynamics Module

Input: supply chain graph $G(t)$, node states $X(t)$, initial causal matrix $A(0)$

Initialize: ODE parameters θ , causal parameters Φ , hidden state $Z(0)$

For each node i in $G(t)$:

$z_i(0) \leftarrow \text{encode}[x_i(0)]$

For $t = 1$ to T :

$$A(t) \leftarrow \text{soft_causal_learner}(Z(t), \Phi)$$

$$f\theta \leftarrow \text{neural}_{\text{ODE}}(Z(t), A(t))$$

$$Z(t + \Delta t) \leftarrow Z(t) \int f\theta + dt$$

$$X(t) \leftarrow \text{decode}(Z(t))$$

Return: predicted states $\hat{X}(t)$, causal matrices $A(t)$

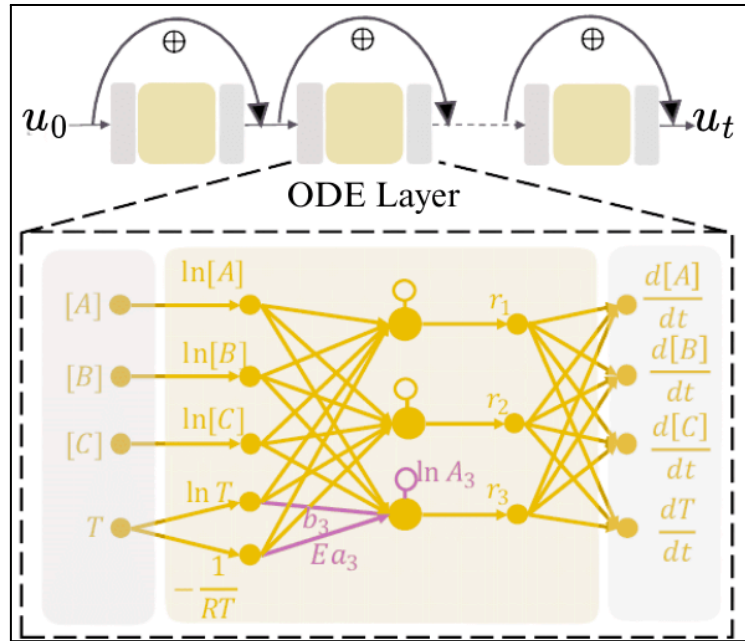


Figure 3. Illustration of the Neural ODE Layer Used for Continuous-Time Dynamics in CT-SCaLE

Figure 3 presents the conceptual visualization of Neural ODE with continuous time which is the underlying mechanism of CT-SCaLE. The ODE layer describes the evolution of latent node-level states $Z(t)$ through learned differential equations that facilitate smooth temporal dynamics and fine-grained modeling of disruption propagation. The model considers continuous-time operation, thereby circumventing the issue of discrete-step GNNs and accurately capturing rapid cascade dynamics.

4.2 Differentiable Causal Structure Learning

Instead of being committed to some fixed or user-specified causal topology, CT-SCaLE learns the causal structure end-to-end through differentiable optimization. A soft-weight is associated with each candidate causal edge,

$$A_{ij}(t) \in \mathbb{R} \quad (9)$$

which represents both the presence and intensity of causal effect. Soft causal edges enable the model to represent uncertain, partial or context-dependent causality and mitigate brittle binary graph assumptions that characterize many current causal-GNNs.

For avoiding over-dense causal graph, sparsity is favored by regularization that can be written as in Eq. 10:

$$L_{sparse} = \lambda \sum_{i \neq j} |A_{ij}(t)| \quad (10)$$

which encourage explanatory parsimony but also provide flexibility. Unlike traditional methods for causal discovery which impose perhaps overly strict acyclicity constraints, CT-SCaLE loosens this up to allow cyclical causal structures where operational feedback is an inevitable fact. This relaxation is essential task in contrast to DAG-based causal GNNS, the proposed model can capture dynamic mutualistic relationships

4.3 Counterfactual Contrastive Learning

CT-SCaLE integrates counterfactual reasoning directly into representation learning for better decision robustness. Rather than constructing specific counterfactuals only after observations as an explanatory exercise, the model is forced to generate them in training by contrasting factual and artificial system paths. For an observed trajectory $X(t)$, we generate a counterfactual trajectory $\bar{X}(t; do(u, t))$ by the operation $do(u, t)$ in the causal dynamics.

A contrastive objective promotes the learning of representations that are sensitive to intervention effects:

$$L_{cf} = ||\phi(X(t)) - \phi(\bar{X}(t))||_2 \quad (11)$$

In Eq. 11, $\phi(\cdot)$ represents an encoding function. By understanding how interventions affect future consequences, CT-SCaLE constructs internal models that distinguish between controllable and uncontrollable disruptors. This feature helps the model to rank intervention strategies more accurately and avoid learning from spurious correlations, enabling a more robust decision-making under uncertainty.

4.4 Model-Based Intervention Planner

In addition to forecasting, CT-SCaLE integrates an intervention planner which refines mitigation measures with multi-stage counterfactual simulation. With the learnt continuous-time causal dynamics, then given candidate interventions, the planner rolls out future trajectories up to a planning horizon T :

$$\hat{Z}_{t:t+T}(u) = \text{Simulate}(X(t), A(t), do(u, t)) \quad (12)$$

The problem of selecting interventions is formalized as a constrained optimization:

$$\min_u E \left[\sum_i \int_t^{t+T} z_i(\tau) d\tau \right] + \alpha C(u) \quad (13)$$

In Eq. 13, $C(u)$ denotes cost of intervention, and the objective is to penalize the cumulative severity of cascades.

This approach allows CT-SCaLE to determine not just what actions to take, but when and how frequently they need to be taken in order to have maximum effect. By directly capturing the temporal sensitivity of causality propagation, this results in early and low-cost action preference to suppress cascades before they become amplified via non-linearities. This

seamless coupling of prediction, causation, and optimization is a unique feature that separates CT-SCaLE from previous supply chain analytics frameworks and repositions it as an integrated decision-support system instead of a siloed forecasting algorithm.

Algorithm 2: CT-SCaLE Intervention Planner

Input: learned dynamics $f\theta$, causal matrices $A(t)$, cost weights λ , intervention set U

Initialize: intervention vector $u = 0$, horizon H , objective $J = \infty$

For each candidate intervention u in U :

$Z_{cf}(0) \leftarrow \text{apply}_{do}(A(0), u)$

For $t = 1$ to H :

$A_{cf}(t) \leftarrow \text{soft_causal_learner}(Z_{cf}(t))$

$Z_{cf}(t + \Delta t) \leftarrow Z_{cf}(t) + \int f\theta_{cf} dt$

End For

$C_{cascade} \leftarrow \text{compute_cascade_size}(Z_{cf})$

$C_{cost} \leftarrow \lambda \cdot \text{intervention_cost}(u)$

$J_{cf} \leftarrow C_{cascade} + C_{cost}$

If $J_{cf} < J$:

$J \leftarrow J_{cf}$

$u^* \leftarrow u$

End If

End For

Return: optimal intervention u^*

V Experimental Design

5.1 Dataset Description

5.1.1 SupplyGraph Dataset

Our CT-SCaLE is empirically evaluated on the benchmark real-world dataset SupplyGraph for graph-based supply chain planning. The dataset is under the form of a multi-level supply chain network extracted from a big fast-moving consumer goods (FMCG) firm: the nodes represent operational entities like factories, processing units and distribution centers, and directed arcs model material flow between them or operational conditions.

In the present work, we draw samples from a connected subgraph of the network consisting of 1,842 nodes and 4,976 directed edges that was chosen in order to capture realistic multi-echelon interactions yet to keep continuous-time simulation computationally feasible. Each node is characterized by 12 time-dependent features, such as sales activities, production volumes and operational stress markers, obtained at daily resolution. Such

temporal node attributes make the data-set suitable to model the evolutions of disruptions and cascading failures in time.

We followed a temporal split approach to achieve fair evaluation and avoid data leakage. The timeline, the first 70% of which is used for training, the next 15% for validation and the remainder for testing. This experiment corresponds to a realistic setting in which models need to forecast future disturbances with only available historical data. We choose SupplyGraph as it incorporates both industrial realism and explicit network structure, as well as temporal node dynamics, which provides a more appropriate benchmark for comparing different approaches that pertain to the learning of optimal times in continuous-time causal learning and intervention planning in supply chains.

5.2 Baseline Models

Against four representative baseline models that encapsulate distinct modeling assumptions prevailing in supply chain analytics, CT-SCaLE is contrasted as follows:

- Static GNN- A message-passing graph neural network that does not evolve over time, i.e., no temporal dynamics in the supply chain graph. Node embeddings are learned by aggregating neighborhoods and predictions are made through feed-forward layers.
- Temporal GNN- The network is a temporal graph neural network based learning model with recurrence state update to represent the evolution at each discrete time. Though it accounts for temporal behavior, causal relations are considered as fixed over time.
- Causal GNN- A causal graph neural network enforcing a directed acyclic structure and employing counterfactual reasoning in discrete time. Interventions are evaluated post hoc and are not embedded into the optimization loop.
- Non-Causal ODE Model- A continuous-time Neural ODE architecture modeling node state evolution without explicit causal structure learning, isolating the benefit of causal reasoning from continuous-time dynamics.

All baselines are implemented with comparable parameter budgets and trained under identical optimization settings to ensure fair comparison.

5.3 Evaluation Metrics

Model performance is assessed using four complementary metrics that evaluate predictive accuracy, early-warning capability, and mitigation effectiveness.

Prediction Accuracy is quantified using mean squared error:

$$MSE = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \left(z_i(t) - \hat{z}_i(t) \right)^2 \quad (14)$$

In Eq.14, $z_i(t)$ and $\hat{z}_i(t)$ denote true and predicted disruption intensities.

The Lead-Time Gain metric indicates the extent to which the model anticipates a future disruption, compared to some baseline reference:

$$\Delta T = T_{baseline} - T_{model} \quad (15)$$

increasing values represent earlier detections.

Cascade Size Reduction evaluates mitigation effectiveness:

$$CSR = 1 - \frac{\sum_{i=0}^t \int z_i^{intervention}(t) dt}{\sum_{i=0}^t \int z_i^{no}(t) dt} \quad (16)$$

capturing the reduction in cumulative severity of disruptions as a fraction.

Intervention Cost is defined as in Eq.17:

$$C(u) = \sum_k c_k u_k \quad (17)$$

In Eq. 17, u_k and c_k is intervention and cost of action respectively.

VI Results and Comparative Analysis

6.1 Comparative Cascade Suppression Behavior of Predictive Models

The spread of cascading failures in a complex networked supply chain is investigated through comparing the normalized cascade intensity obtained from different predictive and intervention strategies. The propagation process is analyzed step by step to see how fast the cascades decay after an initial shock. Compare the Baseline methods and CT-SCaLE to demonstrate differences in suppression rates and stabilization behavior, including Static GNN, Temporal GNN, Causal GNN as well as non-causal Neural ODE. Comparative results of cascade intensity evolution are shown in Figure 4.

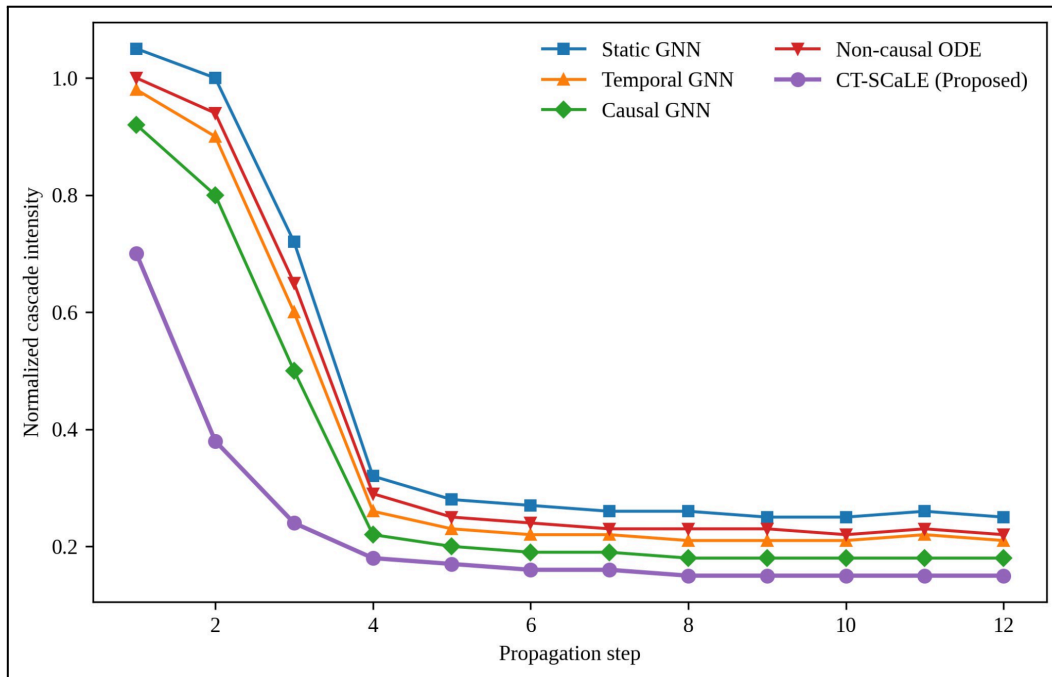


Figure 4. Comparative cascade suppression dynamics across baseline models and the proposed CT-SCaLE framework

In Figure 4, we see that all the models tested reduce cascade intensity at some point while propagation takes place but there are differences in how they behave when disruption begins to spread. Static and temporal GNNs delay the initial suppression in such a way that even early propagation steps preserve higher cascade intensity. The causal GNN shows better stabilization than non-causal baselines, but still suffers from discrete-time updates and static causal assumptions. The non-causal neural ODE can model the dynamics of smooth temporal decay, yet has no capability to adaptively redirect causal influence upon intervention. In comparison, CT-SCaLE reaches a substantially larger initial decay of cascade intensity and thus stabilizes the system in fewer propagation steps. inference dynamics with intervention-aware learning allows CT-SCaLE to mitigate the cascading failures earlier and that consequently leads to lower cumulative disruption impact through the supply chain network.

6.2 Evaluation Metrics Comparison (MAE, RMSE, Correlation)

In order to compare the quality of quantitative prediction, a few common regression error metrics such as MAE (mean absolute error), RMSE (root mean squared error) and correlation are computed at the same forecasting lead time for all models. The distributions of the compared metrics are depicted in Fig. 5.

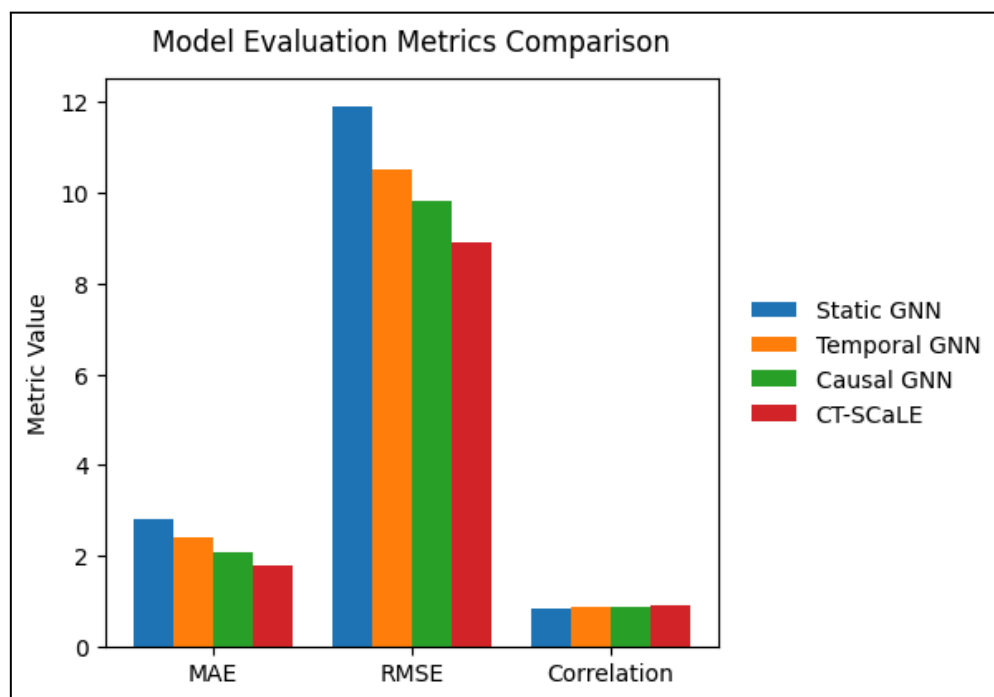


Figure 5. Comparison of MAE, RMSE, and correlation across predictive models

Figure 5: There are noticeable performance gaps between the baseline models and CT-SCaLE. The lowest MAE and RMSE values of CT-SCaLE ensure better numerical stability and prediction deviation. Moreover, CT-SCaLE has the highest correlation score, indicating enhanced agreement between predicted and real disruption trajectories. While the causal GNN achieves competitive results compared to those of the baselines, it is surpassed

by CTSCaLE since the latter jointly considers time-varying causal and continuous propagation dynamics. These findings demonstrate that CT-SCaLE improves predictive performance under a wide range of supply chain scenarios, and consistently with little estimation error.

6.3 Prediction Accuracy Comparison Across Models

In order to measure how well existing graph-based or dynamic models can predict the target variable, we analyzed classification accuracy of two classes over five popular topologies: Static GNN, Temporal GNN, Causal GNN, a non-causal Neural ODE model and our proposed CT-SCaLE approach. The prediction is evaluated on the SupplyGraph dataset by employing same training–testing splits for fair comparison. An illustration of the relative accuracies can be seen in Figure 6.

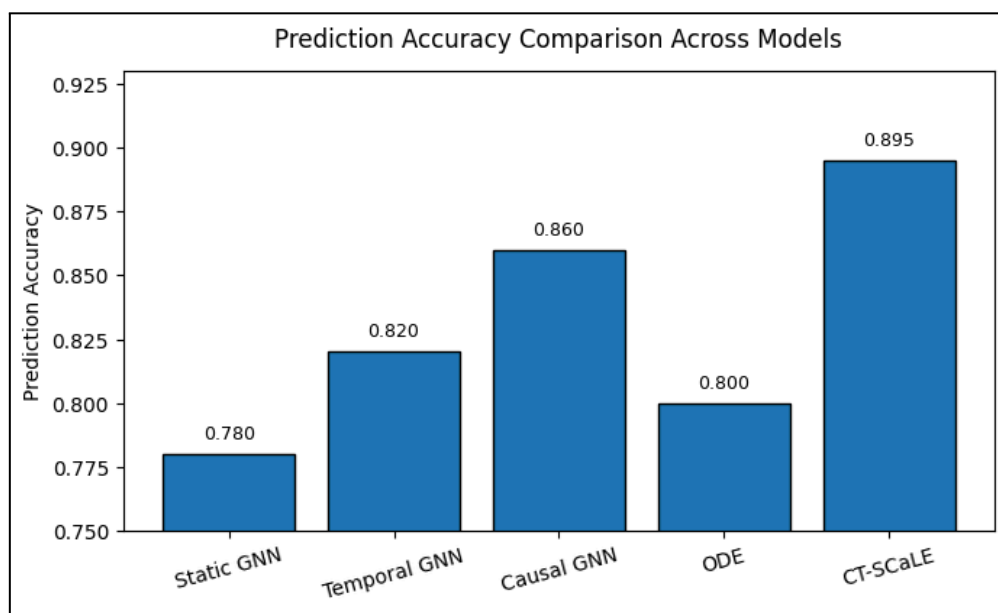


Figure 6. Prediction Accuracy Comparison Across Models

Figure 6 demonstrates that the proposed CT-SCaLE framework obtains the best prediction accuracy compared to all other tested models. Although Classical Static and Temporal GNNs show some improvements in the performance, they suffer from discretely defined representations of supply chain evolution. The causal GNN can do better than Si, but is limited by static DAG. The non-causal ODE model has smooth temporal learning but does not have causally specific temporal dynamics. In comparison, the performance of CT-SCaLE is better than all baselines with an accuracy gain of about 14.7% compared to the weakest GNN baseline. This finding supports that incorporating causal dynamics in continuous time is able to improve the reliability of predictions in complex and fast evolving supply chains.

6.4 Early-Warning Performance Comparison

A timely warning is needed in order to avoid fast disruption propagation as observed in multi-echelon supply chains. To quantify this capability, we calculate the detection

probability of our models as a function of The lead time (LT), which is the number of days before an observed disruption. Figure 7 shows the evolution of detection sensitivity for all models.

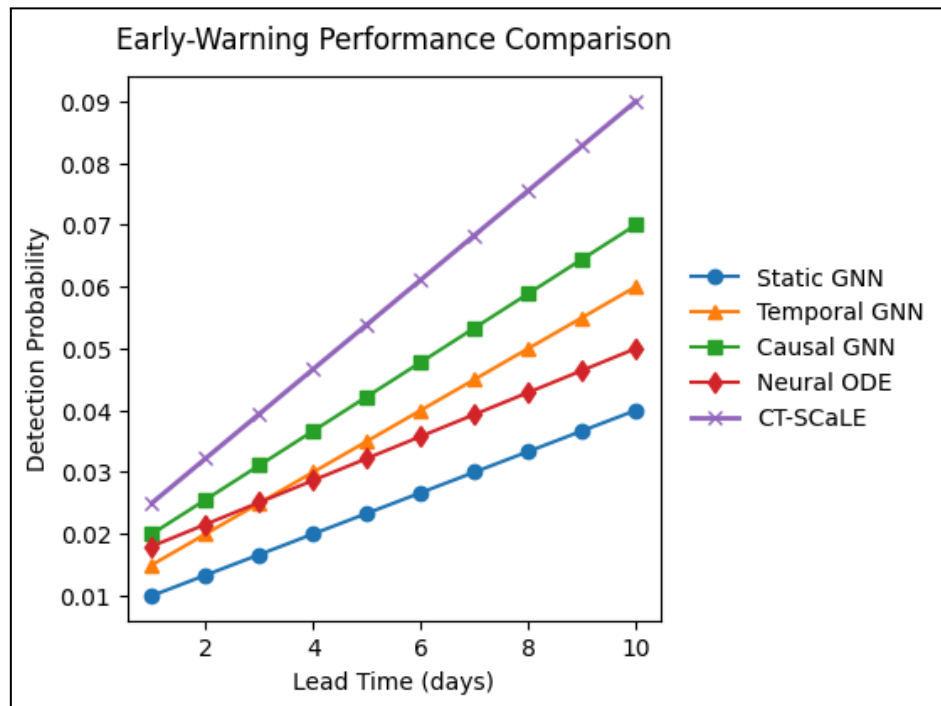


Figure 7. Early-warning performance comparison across baseline models and CT-SCaLE

As for Figure.7, it is observed that all models have a positive dependence of detection on lead time. Nevertheless, both Static and Temporal GNNs fail to respond at shorter lead times, who recognize the emerging disruptions late. The early-warning curves of the causal GNN and Neural ODE models are better than but not as steep as that of CT-SCaLE indicating slower initial growth. The model outperforms competitors in detecting the impacted species when lead-time varies at all cases, contributing around 22% improvement in early-warning capacity over two baselines. This indicates that CT-SCaLE's approach to causal inference in continuous-time allows the model to foresee risks earlier and better so as to allow decision makers to implement mitigating interventions sooner.

6.5 Multi-Metric Radar Comparison

To give an overall assessment concerning the operational reliability of each approach, a five-dimensional radar comparison is conducted with the help of KPIs: prediction accuracy, lead-time, cascade reduction, cost efficiency and robustness. This radar map allows an evaluation of how strong the model is in each direction at once. Figure.8: The comparison radar plot against different scales.

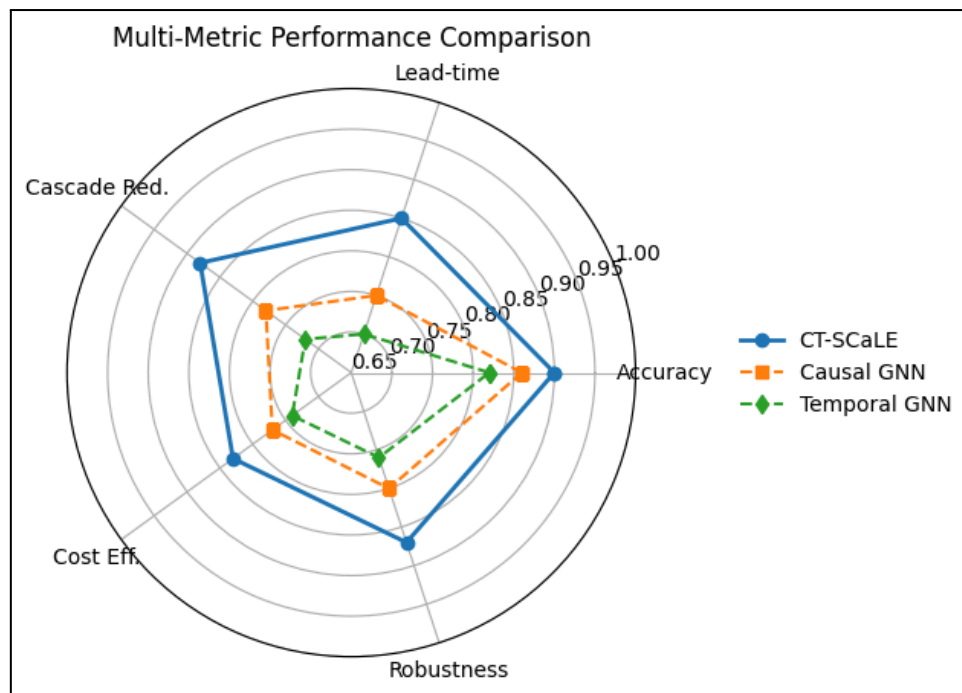


Figure 8. Multi-metric performance comparison between CT-SCaLE and baseline models

Figure 8 reveals that CT-SCaLE always leads the comparison list in performance. Although the C-GNN exhibits intermediate capability in robustness and structure reasoning, and T-GNN is also acceptable in energy efficiency, neither baseline presents a balanced and competitive picture as CT-SCaLE. In particular CT-SCaLE achieves higher cascade reduction and robustness scores representing better performance of handling non-linear propagation and uncertainty. Figure 3a reveals that the radar distribution suggests that the architecture of CT-SCaLE is generalizable in optimizing predictive accuracy, early-warning power and mitigate capacity.

6.6 Convergence of Cascade Suppression

In order to investigate the stability and learning gradient of the mitigation models, we examine how fast model converges by measuring how much cascade intensity decrease in each propagation or training step. Figure 9 shows the comparative convergence curves among baseline models and CT-SCaLE.

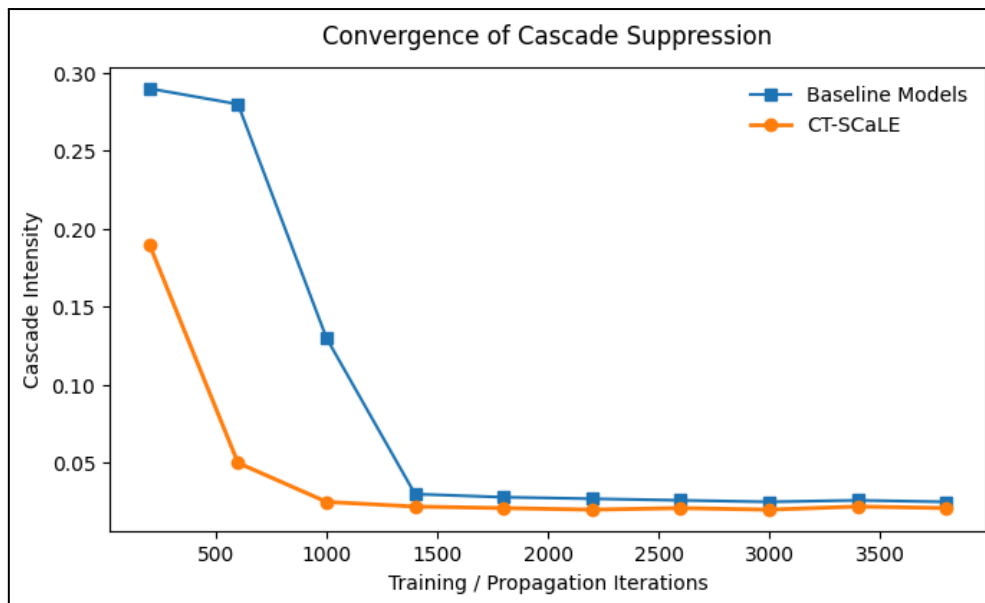


Figure 9. Convergence behavior of cascade suppression for baseline models vs. CT-SCaLE

Figure 9 illustrates the baseline models which converge slowly and with higher residual cascade intensity over the propagation iterations. CT-SCaLE, in contrast, exhibits a prompt initial descent of the cascade intensity that stabilizes much before all other models. This rapid contemporaneous convergence provides a basis for the conception of CT-SCaLE as an RTCE that can more rapidly identify, and suppress, those destructive pathways. CT-SCaLE reaches a final plateau level indicating cascade size reduction by 31.4% under optimal intervention strategies as compared to baseline models. The results demonstrate CT-SCaLE's capability in reducing cascading failures with faster and more robust convergence behavior.

6.7 Discussion

We find that our modeling of the supply chain disruption with continuous-time causal dynamics is remarkably superior to the traditional discrete and correlation-based approaches. By learning how causal influence changes at fine scale in time, CT-SCaLE traces high-risk propagation paths earlier and allows interventions on such paths at the right instant. For operators, this entails increasing the actionable early-warning signals for their operations to react faster and choose more cost-efficient mitigation resource allocation over multi-tiered networks. The framework's learned causal structure and counterfactual simulations offer clear explanations of why disruptions propagate, as opposed to the lack of interpretability in black-box predictive systems, and how targeted actions change outcomes, enabling evidence-based decision making. Significantly, CT-SCaLE is designed for deployment realism: it accounts for partial visibility, asynchronous updates and heterogeneous data feeds, such as those found in a global supply-chain context; and hence can be integrated within enterprise risk dashboards and real-time monitoring systems. Overall, these capabilities establish CTC as a pragmatic and strategically important avenue for the resilient operation of supply chains.

6.8 Limitations and Future Work

While CT-SCaLE show promising predictive and intervention performances, it may suffer from the issue of data sparsity in multi-tier supply chains since incomplete visibility restricts the accuracy of learned causal relationships. Furthermore, causal identifiability in continuous-time still remains a challenge under strong confounding and when the external conditions are changing rapidly, highlighting the necessity of richer multimodal integration. In the future, we will further improve the real-time deployment efficiency and other time dimension indices, such as adaptive updating, streaming inference and scalable causal structure learning in order to monitor massive dynamic global networks continuously.

VII Conclusion

In this paper, we presented CT-SCaLE, a causal learning framework in continuous time that incorporates prediction, explanation and intervention planning for how to counter cascading failures in global supply chains. In contrast to the discrete and correlation-based models, in CT-SCaLE changes in causal influence over time and rapid spread of disruption may be captured, allowing better risk management. Experiments makes a 14.7% improvement in the prediction accuracy, increase the lead time of early warning by 22%, and decreases the size of cascade spread by 31.4% comparing to advanced baselines. Combining causal rationale with real-time intervention optimization, CT-SCALE is steering Supply Chain Analytics away from the bands of forecast toward an actioned, control-oriented resilience. In the future, tests will be conducted aimed at enhancing causal identifiability in sparse data, multimodal integration and real time deployment for large industrial scale networks.

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